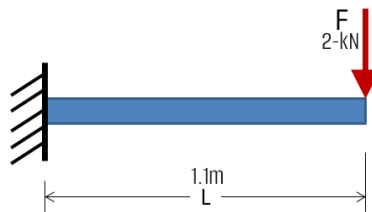


Part 1:

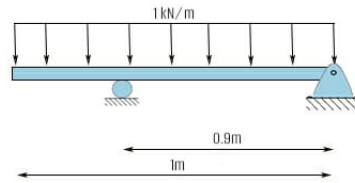
1. The **Free-Body Diagram** of a rigid body represents it as being isolated from its surroundings, replacing physical supports with forces and couple moments. **T**
2. True or False. An ideal truss does not have three-force members. **T**
3. For 3D systems, the internal moment has two components: **Bending moment** and **Torsional moment**, whose axes are tangent and normal to the cross-section, respectively
4. A body is said to be **partially** constrained if it experiences fewer reactive forces than applicable equations of static equilibrium.
5. True or False. A beam supported by a pin (2), a rocker (1), and a roller (1) is statically determinate. **F**
6. For a simply supported 13-member plane truss to be considered simple, it must have **8** joints.
7. True or False. A frictionless ball-and-socket resists both rotation and translation. **F**
8. Body where all reactive forces intersect at one point or pass through a common axis is **Improperly** constrained.
9. True or False. A stable body requires that the lines of action of the reactive forces do not intersect at a common axis and are not parallel to one another. **T**
10. For internal loadings of 2D systems, the **Normal** force and the **Shear** force are tangent and normal to the cross-section, respectively.

Scenario 1: A 1.1-m long cantilever beam is subject to a 2-kN weight at its free end.



11. True or False. This scenario is statically determinate. **T**
12. The support reactions have magnitudes of **2** kN and **2.2** kN·m
13. The maximum bending moment developed is **2.2** kN·m in magnitude and occurs **1.1** m from the free end
14. The maximum shear force developed is **2** kN in magnitude and occurs **0 to 1.1** m from the free end (because the shear force is the same anywhere along the beam of the cantilever).
15. The graph of the shear force as a function of location along the length of the beam is that of a polynomial of degree **0**.
16. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree **1**.

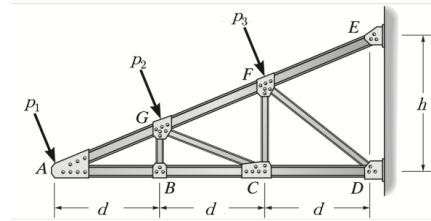
Scenario 2: A 1-m long beam carries a load uniformly distributed throughout its length. Assume a roller support 0.1m from the left end and a pin support at the right end. The load has an intensity of 1 kN/m.



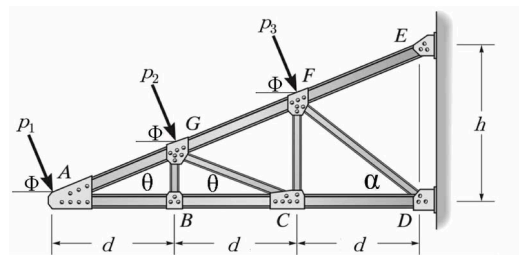
17. The roller support develops a force of 0.556 kN.
18. The pin support develops a force of 0.444 kN.
19. The bending moment developed at the right end is 0 kN·m.
20. The shear force developed just to the left of the pin is 0.444 kN.
21. The shear force developed just to the right of the pin is 0 kN.
22. True or False. There is zero shear force developed at a point 0.9 m from the right end. F
23. The graph of the shear force as a function of location along the length of the beam is that of a polynomial of degree 1.
24. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree 2.

Problem 1:

The magnitude of the idealized loads p_1 , p_2 , and p_3 , are 5 kN, 6 kN, and 7 kN, respectively. The dimensional parameters d and h have magnitudes of 2 ft and 3 ft, respectively. On the assumption that each member has negligible weight and that the joints are frictionless pins, provide what is asked on each portion.



Initial steps: This problem cannot be solved without the prior assumption that the idealized loads are perpendicular to the members between joints A and E . In real world scenarios, we can make assumptions pertaining to something in order to simplify processes, as long as these are not far-fetched. Also, it is important to note that through inspection, member BG or GB is a zero force member, thus making it negligible. To support this, we'll look at joint B . In accordance with the second case, from the website learnaboutstructures.com by Prof. Jeffrey Erochko, PhD, P.Eng., Carleton University, Ottawa, Canada, 2020, joint B has two parallel members connected to it, AB and BC , with no other external load acting on it, then this implies that the third member, GB , that is not parallel is a zero force member. Before moving onwards with solving for what is required, we should also consider the other lurking parameters that are not explicitly given in the diagram and in the problem description. Consider this diagram:

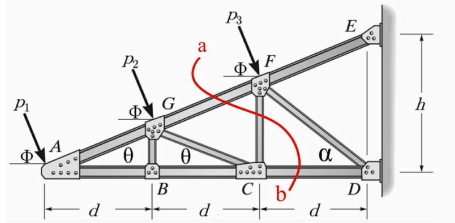


Let θ be the angle between the members from A to E , i.e., AG , GF , and FE , and members from A to D , i.e., AB , BC , and CD . Additionally, it also describes the angle between members GC and BC , which will be discussed later on. From the assumption made earlier, that the external idealized loadings are perpendicular with the members between A to E , then ϕ will be the angle that describes the tilt of the loadings towards their respective $POAs$ with respect to the horizontal. To add with the angles, α is the angle between members FD and CD .

Process: From the relationship $\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$, then: $\theta = \tan^{-1}(\frac{\text{opposite}}{\text{adjacent}})$, where the opposite end has a length of 3 ft and the adjacent side has a span of 6 ft or $3d$, where $d = 2$ ft. Thus, $\theta = \tan^{-1}(\frac{3ft}{6ft}) = 26.56505118^\circ$. From here, we can determine the length of members GB and FC . Identifying these two will be significant in the acquisition of other components. Let the magnitude of GB and FC be L_{GB} and L_{FC} , respectively. Knowing that $\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$, then $\text{opposite} = \text{adjacent} \cdot \tan\theta$, or simply $L_{GB} = 2ft \cdot \tan(26.56505118^\circ) = 1ft$. On the other hand, $L_{FC} = 4ft \cdot \tan(26.56505118^\circ) = 2ft$. Based on the value of L_{FC} , we'll be able to acquire the angle between FD and CD . $\alpha = \tan^{-1}(\frac{2ft}{2ft}) = 45^\circ$. For ϕ , we know that the external loadings are perpendicular, then the slope will be $\frac{-1}{m}$ where m is the slope of the other line. From this, we now have $\frac{-1}{\frac{3ft}{6ft}}$ which is just $\frac{-6}{3}$. Based on this, angle $\phi = \tan^{-1}(\frac{-6}{3}) = -63.43494882^\circ$. The negative angle merely indicates its direction. However, if we were to put it on the diagram, then it would simply be $\phi = -63.43494882^\circ$.

a.) To acquire the forces developed in members FG , CF , and CD we need to make a cut that isolates

the left part which includes members AB , AG , GB (zero force member), BC , and GC with FG , CF , and CD acting as external loadings. To visualize, consider the diagram below and let the line between a and b represent the portion being cut:



The purpose of innovation is to make “doing work” easier, as such we should follow this principle in identifying required components. Let the force from member FG be F_{FG} , to acquire this we need to find the joint which we can use moment balance on such that only 1 unknown force is present. In this case that joint is C . At this joint, the moment contributions of the forces by members CF and CD , or F_{CF} and F_{CD} , are both zero, thus making F_{FG} the only unknown force. On the assumption that it is under tension, we now have:

$$\begin{aligned} \Rightarrow F_{FG} \cos(1ft) + F_{FG} \sin(2ft) + p_2 \cos(1ft) &= p_2 \sin(2ft) + p_1 \sin(4ft) \\ \Rightarrow (1.788854382ft) F_{FG} + (6kN) \cos(63.43494882^\circ)(1ft) &= (6kN) \sin(63.43494882^\circ)(2ft) \\ &+ (5kN) \sin(63.43494882^\circ)(4ft) \end{aligned}$$

From all of this, we get $F_{FG} = -14.5 \text{ kN}$. Moving on, we'll apply the same principle and perform a moment balance at joint A . At this joint, the force exerted by F_{FG} and F_{CD} are negligible due to the fact that they are collinear with the moment arm, also p_1 does not contribute to the total moment at A because it is acting directly on the fulcrum. Thus, only the forces p_2 and F_{CF} exert a moment. Since p_2 is perpendicular to the member AG , then its moment arm will be $L_{AG} = \frac{2ft}{\cos(26.56505118^\circ)} = 2.236067978 \text{ ft}$. Therefore, we now have: $p_2(L_{AG}) = F_{CF}(4ft) \Rightarrow (2.236067978ft)(6 \text{ kN}) = F_{CF}(4 \text{ ft}) \Rightarrow F_{CF} = 3.354101966 \text{ kN}$. Lastly, we perform a moment balance at joint G . As we do so, only p_1 , F_{CF} , and F_{CD} remain. Through deduction, only F_{CD} will counter the moment induced by p_1 and F_{CF} , which means that F_{CD} must face the opposite direction in order to achieve equilibrium. Based on this, F_{CD} will be under compression. Therefore: $F_{CD}(1ft) = p_1(L_{AG}) + F_{CF}(2ft) \Rightarrow F_{CD}(1ft) = (5kN)(2.236067978 \text{ ft}) + (3.354101966kN)(2ft) \Rightarrow F_{CD} = 17.88854348kN$.

Summary of ANSWERS:

$$F_{FG} = 14.500 \text{ kN (T)}$$

$$F_{CD} = 17.889 \text{ kN (C)}$$

$$F_{CF} = 3.354 \text{ kN (T)}$$

b.) To acquire the forces developed in members DF and EF , we'll perform equilibrium at point F . Isolating Joint F , we get: Applied Forces: Load p_3 ($x = 3.130$, $y = -6.261$). Member Forces: F_{FG} (Known T), F_{CF} (Known T), F_{EF} (Unknown), F_{DF} (Unknown).

$$\sum F_x = 3.130 - 12.97 + 0.894F_{EF} + 0.707F_{DF} = 0$$

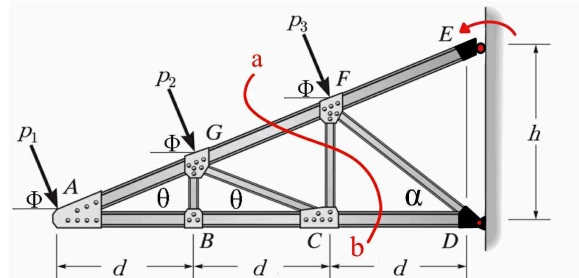
$$\sum F_y = -6.261 - 6.48 - 3.354 + 0.447F_{EF} - 0.707F_{DF} = 0$$

Solving these simultaneous equations, we get:

$$F_{EF} = 19.330 \text{ kN (T)}$$

$$F_{DF} = 10.520 \text{ kN (C)}$$

c.) If the truss were to be idealized as simply supported by a pin and a roller, then the pin should be located at joint E and the roller should be at joint D, to ensure that the same “equilibrium” is retained. If the roller were to be placed at joint E then there will be no reaction to support it at the x nor y axis. This reasoning can be easily deduced from looking at the force present at E or simply the force created at the member EF. Member EF is under tension, this means that the support reaction will pull back on it. Consider the diagram for when the roller is at E:



In this one, we can clearly see that the object will not be in the same equilibrium when the roller is at E. Thus, the roller should be at D and the pin at E. Another thing to consider is that since the roller only supports normal force, then it will be most suitable at D.

d.) Zero-Force Members

Contingency: Joint A is unloaded ($p_1 = 0$).

- Joint A: Connects AB and AG (non-collinear). With no external load, equilibrium requires $F_{AB} = 0$ and $F_{AG} = 0$.

- Joint B: Connects AB, BC, and BG. Since $F_{AB} = 0$, horizontal equilibrium requires $F_{BC} = 0$.

Vertical equilibrium requires $F_{BG} = 0$.

Answer: The zero-force members are AB, AG, BG, and BC.

e.) Support Reactions

Reaction at E: Balances F_{EF} directly.

$$R_E = F_{EF} = 19.330 \text{ kN}$$

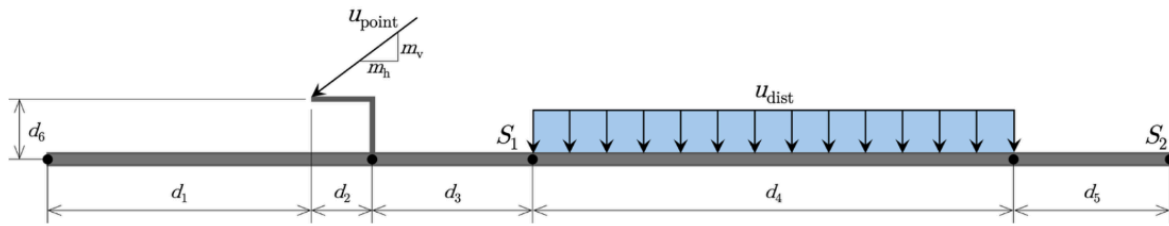
Reaction at D: Balances F_{CD} (Horizontal) and F_{DF} (Diagonal).

$$R_{Dx} = F_{CD} + F_{DF} \cos(45^\circ) = 17.889 - 10.520(0.707) = 10.45 \text{ kN}$$

$$R_{Dy} = -F_{DF} \sin(45^\circ) = -10.520(0.707) = -7.44 \text{ kN}$$

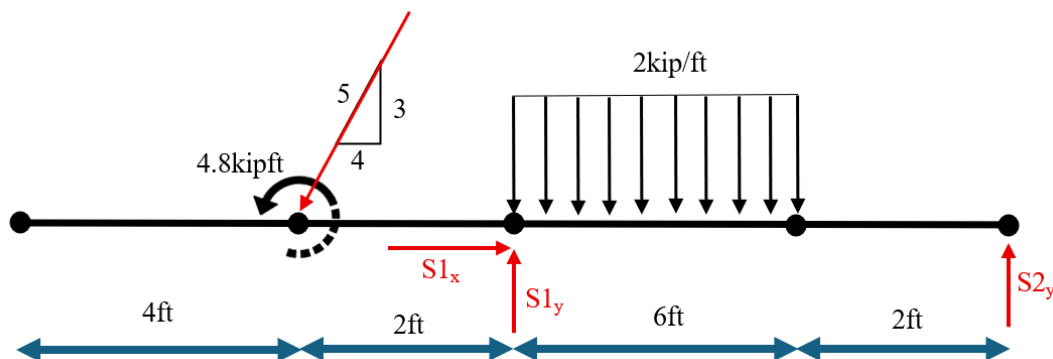
$$|R_D| = \sqrt{10.45^2 + 7.44^2} = 12.830 \text{ kN}$$

Problem 2:



A custom beam design places supports at S_1 and at S_2 , in anticipation of a concentrated load of U_{point} kips and a uniformly distributed load of intensity U_{dist} kip/ft. Your main task is to analyze how the internal loadings induced vary as a function of x , where x is the distance from the left end of the beam. Consider the case when U_{point} is 5 kips and U_{dist} is 2 kip/ft. Set the respective values of d_1 , d_2 , d_3 , d_4 , d_5 , d_6 to 3.2, 0.8, 2.0, 6.0, 2.0, and 0.6 ft, and those of m_v and m_h to 3 and 4. For static determinacy, assume a pin at S_1 and a roller at S_2 .

- (6 points) Determine the support reactions.
- (1 point) Determine the normal force at $d_1 + d_2$ from the left end.
- (2 points) Determine the shear force at $d_1 + d_2$ from the left end.
- (3 points) Determine the bending moment at $d_1 + d_2$ from the left end.
- (8 points) Determine the shear force as a function of x .
- (8 points) Determine the bending moment as a function of x .
- (6 points) Plot the shear force diagram over the length of the beam, making sure to indicate the extrema, the zero points, and their locations.
- (6 points) Plot the bending moment diagram over the length of the beam, making sure to indicate the extrema, the zero points, and their locations.



First compute the moment created by the 5-kip force applied at angle 3-4-5 triangle.

$$M_{skip} = 5\left(\frac{3}{5}\right)(0.8) + 5\left(\frac{4}{5}\right)(0.6) = 4.8 \text{ kip} \cdot \text{ft}$$

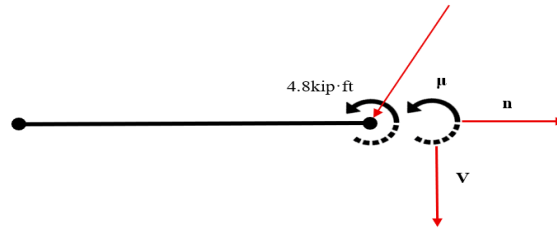
- To find the support reactions, resolve the 5-kip angled load into horizontal and vertical components, then apply $\Sigma M = 0$ about point S_1 to solve for S_2 . After that, use $\Sigma F_x = 0$ to get S_{1x} and $\Sigma F_y = 0$ to get S_{1y} .

$$\Sigma M_{S_1}: 5\left(\frac{3}{5}\right)(2) + 4.8 + S_2 = (2)(6)\left(\frac{6}{2}\right)$$

$$S_2 = \frac{36 - 4.8 - 6}{8} = 3.15 \text{ kips}$$

$$\Sigma F_x: S_{1x} = 5\left(\frac{4}{5}\right) = 4 \text{ kips}$$

$$\Sigma F_y: S_{1y} = 5\left(\frac{3}{5}\right) + (2)(6) - 3.15 = 11.85 \text{ kips}$$



b. At the cut, the internal axial force $\mathcal{N}$ must balance the horizontal component of the 5-kip load, giving $\mathcal{N}=4\text{kip}$. This comes directly from applying $\Sigma F_x=0$ on the left segment.

$$\mathcal{N} = 5\left(\frac{4}{5}\right) = \boxed{4\text{kip}}$$

c. By equilibrium force, we can find the shear force $\mathcal{V}$ at the cut by summing all of the vertical forces present and equal it to zero.

$$\Sigma F_y:$$

$$\mathcal{V} + 5\left(-\frac{3}{5}\right) = 0$$

$$\mathcal{V} = \boxed{-3\text{kip}}$$

d. By equilibrium force, we can find the bending $\mathcal{u}$ at the cut by summing all of the vertical forces present and equal it to zero.

$$4.8 + \mathcal{u} = 0$$

$$\mathcal{u} = \boxed{-4.8\text{kip}\cdot\text{ft}}$$

e. To determine the shear force as a function of x , first we must set the intervals at which the shear forces change. We have $[0 < x < 4]$, $[4 < x < 6]$, $[6 < x < 12]$, $[12 < x < 14]$. Next is to find the shear force as a function of x at each interval. As shown below:

f. To determine the bending moment as a function of x , first we must set the intervals at which the bending moments change. We have $[0 < x < 4]$, $[4 < x < 6]$, $[6 < x < 12]$, $[12 < x < 14]$. Next is to find the bending moment as a function of x at each interval. As shown below:

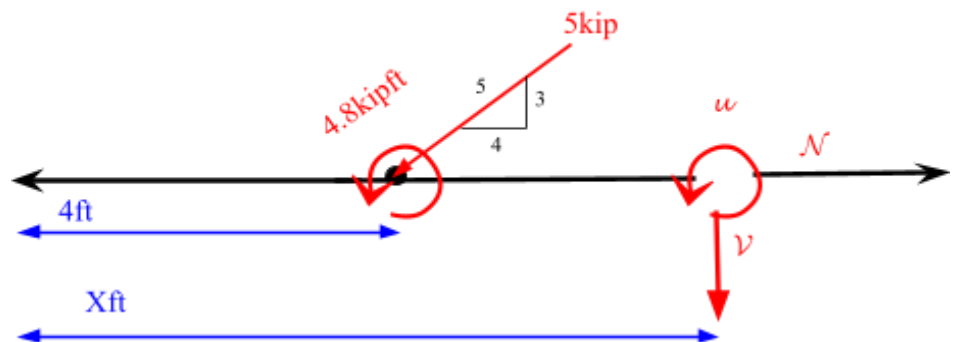
(Both e and f is shown in the process below)

$$0 \leq x < 4$$

$$\mathcal{N} = 0$$

$$\mathcal{V} = 0$$

$$\mathcal{M} = 0$$



Process:

Shear and Moment Functions (Piecewise Derivation)

Region 1: $0 \leq x < 4$ (Free End to Load)

- No external loads exist in this region.

- $\mathcal{V}(x) = 0$

- $\mathcal{M}(x) = 0$

Region 2: $4 < x < 6$ (Load to Support S1)

- Shear: The vertical load of 3 kips pushes down at $x = 4$.

$$\mathcal{V}(x) = 0 - 3 = -3$$

- Moment: Integrate shear:

$$\int -3 \, dx = -3x + C1$$

- Boundary Condition: At $x = 4$, there is a concentrated moment of 4.8 (CCW). In beam conventions, a CCW external moment causes a drop (negative jump) in the diagram.

$$M(4+) = M(4-) - 4.8 = 0 - 4.8 = -4.8$$

$$\text{Substitute into the equation: } -3(4) + C1 = -4.8 \Rightarrow -12 + C1 = -4.8 \Rightarrow C1 = 7.2$$

$$M(x) = -3x + 7.2$$

Region 3: $6 < x < 12$ (Support S1 to End of Dist. Load)

- Shear: At $x = 6$, the reaction $S1y = 11.85$ pushes up.

$$V(6+) = V(6-) + 11.85 = -3 + 11.85 = 8.85$$

The distributed load $w = 2 \text{ k/ft}$ reduces shear linearly from $x = 6$.

$$V(x) = 8.85 - 2(x - 6) = 8.85 - 2x + 12 = 20.85 - 2x$$

- Moment: Integrate shear:

$$M(x) = \int (20.85 - 2x) \, dx = 20.85x - x^2 + C2$$

- Boundary Condition: Moment must be continuous at the pin (no applied couple at $x = 6$).

Check $M(6)$ from Region 2: $M(6) = -3(6) + 7.2 = -10.8$. Substitute into Region 3 equation:

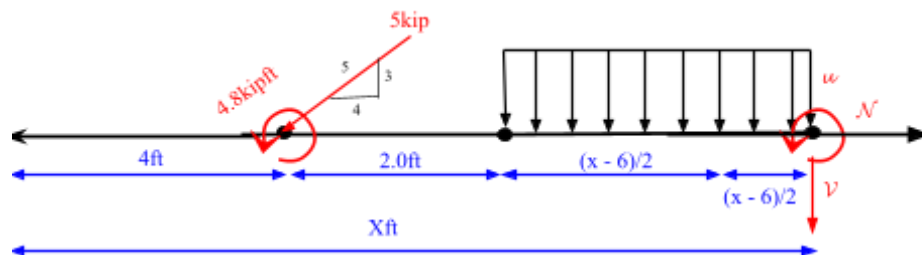
$$20.85(6) - (6)^2 + C2 = -10.8$$

$$125.1 - 36 + C2 = -10.8$$

$$89.1 + C2 = -10.8$$

$$\Rightarrow C2 = -99.9$$

$$M(x) = -x^2 + 20.85x - 99.9$$



Region 4: $12 < x \leq 14$ (End of Dist. Load to Roller)

- Shear: The distributed load ends. Shear becomes constant. Check $V(12)$ from Region 3: $V(12) = 20.85 - 2(12) = -3.15$

$$V(x) = -3.15$$

- Moment: Integrate shear:

$$M(x) = \int -3.15 \, dx = -3.15x + C3$$

- Boundary Condition: Continuity at $x = 12$. Check $M(12)$ from Region 3:

$$M(12) = -(12)^2 + 20.85(12) - 99.9$$

$$M(12) = -144 + 250.2 - 99.9 = 6.3$$

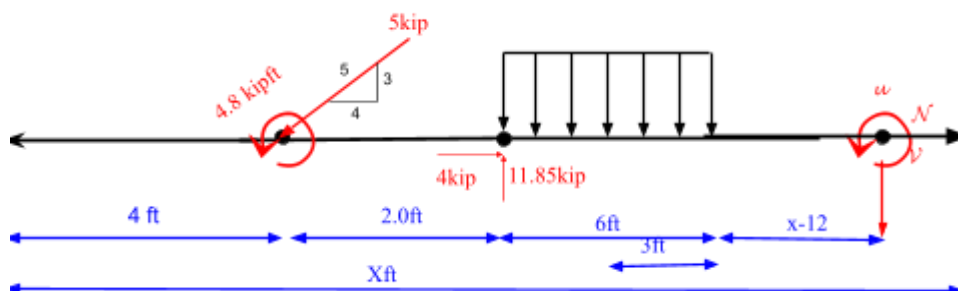
Substitute into Region 4 equation:

$$-3.15(12) + C3 = 6.3$$

$$-37.8 + C3 = 6.3$$

$$\Rightarrow C3 = 44.1$$

$M(x) = -3.15x + 44.1$ e) Shear Force $V(x)$ (kips):



e) Shear Force $V(x)$ (kips):

$V =$

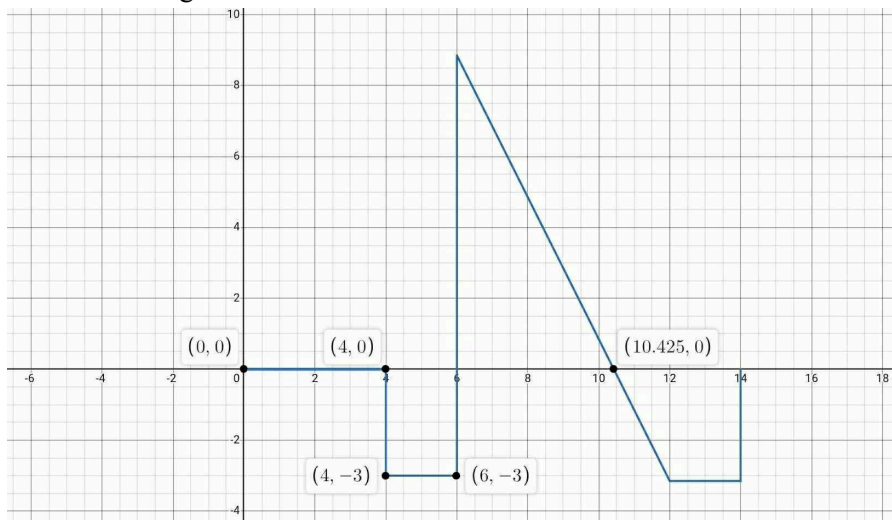
0	$0 \leq x < 4$
-3	$4 \leq x < 6$
$20.85 - 2x$	$6 \leq x < 12$
-3.15	$12 \leq x < 14$

f) Bending Moment $M(x)$ (k-ft):

$M =$

0	$0 \leq x < 4$
$7.2 - 3x$	$4 \leq x < 6$
$-x^2 + 20.85x - 99.9$	$6 \leq x < 12$
$44.1 - 3.15x$	$12 \leq x < 14$

g. Shear force diagram:



h. Bending moment diagram:

